

28.1 Minimum-norm interpolation

Gene expression studies measure tens of thousands of genes on a few hundred patients, and a model with many splines, interactions or random features easily has more columns than rows. This section asks what least squares does when $p > n$. It does not fail, but it no longer picks out one answer, and we study the answer that the SVD and the common algorithms return.

28.1.1 Least squares when p exceeds n

Suppose $\text{rank}(\mathbf{X}) = n < p$, the usual case (a matrix with continuously distributed entries has full rank with probability one). Three consequences follow from Parts I and II.

First, $\mathcal{C}(\mathbf{X}) = \mathbb{R}^n$, so $\mathbf{M} = \mathbf{I}$: every least squares fit reproduces the data, the residuals are zero, $R^2 = 1$, and the estimator s^2 of Theorem 5.6.1 has $n - r = 0$ in its denominator. The data contain no internal information about σ^2 .

Second, $\mathcal{N}(\mathbf{X})$ has dimension $p - n \geq 1$, and the least squares estimates form the affine set $\mathbf{X}^+\mathbf{y} + \mathcal{N}(\mathbf{X})$ (Theorem 10.4.1). Coefficient vectors that differ by an element of $\mathcal{N}(\mathbf{X})$ give the same mean, so the data cannot tell them apart.

Third, $\lambda^\top \beta$ is estimable iff $\lambda \in \mathcal{C}(\mathbf{X}^\top)$ (Theorem 8.2.2), and a coordinate vector $\mathbf{e}_j$ lies in the n -dimensional row space only if it is orthogonal to $\mathcal{N}(\mathbf{X})$, which fails for generic designs. For a generic $\mathbf{X}$ with $p > n$, *no single coefficient is estimable*. Any method that answers questions about individual coefficients is adding information, explicitly or through its algorithm.

Example. (Two interpolators)

Take $n = 4$ observations on $p = 6$ regressors, with the design of the listing below and $\mathbf{y} = (1, 2, 0, -1)^\top$. The minimum-norm solution $\hat{\beta}^+ = \mathbf{X}^+\mathbf{y}$ (Proposition 1.4.3, Proposition 6.10.1) has length 1.322. Adding twice a unit vector of $\mathcal{N}(\mathbf{X})$ gives a second solution of length 2.398. Both reproduce $\mathbf{y}$ exactly. They disagree about the coefficients: the first, for instance, is -0.002 in one and -0.059 in the other. They also disagree about predictions at a point that was not observed: at $\mathbf{x}_0 = \mathbf{1}$ they predict 2.131 and 2.588. The data give no reason to prefer either. The minimum-norm solution is singled out by a choice, not by the data.

```
import numpy as np

rng = np.random.default_rng(2801)
n, p = 4, 6
X = np.round(rng.normal(size=(n, p)), 1)          # a 4 x 6 design, rank 4
y = np.array([1.0, 2.0, 0.0, -1.0])

b_plus = X.T @ np.linalg.solve(X @ X.T, y)        # X^T (X X^T)^{-1} y
print("min-norm solution ", np.round(b_plus, 3))
print("residual norm      ", np.linalg.norm(y - X @ b_plus))

# another interpolator: add a null-space vector of X
_, _, Vt = np.linalg.svd(X)
b_other = b_plus + 2.0 * Vt[-1]                   # Vt[-1] spans part of N(X)
print("other solution     ", np.round(b_other, 3))
```

```

print("residual norm      ", np.linalg.norm(y - X @ b_other))
print("norms", np.linalg.norm(b_plus), np.linalg.norm(b_other))

x0 = np.ones(p) # a new point
print("predictions at x0:", x0 @ b_plus, x0 @ b_other)

```

[Run this code online](#)

28.1.2 The minimum-norm solution

In this chapter the singular values of $\mathbf{X}$ are written $d_1 \geq \dots \geq d_r > 0$, so that σ stays free for the error standard deviation. By Theorem 1.8.1,

$$\mathbf{X} = \sum_{i=1}^r d_i \mathbf{u}_i \mathbf{v}_i^\top, \quad \mathbf{X}^+ = \sum_{i=1}^r d_i^{-1} \mathbf{v}_i \mathbf{u}_i^\top,$$

with orthonormal $\mathbf{u}_1, \dots, \mathbf{u}_r \in \mathbb{R}^n$ and $\mathbf{v}_1, \dots, \mathbf{v}_r \in \mathbb{R}^p$. The matrix

$$\mathbf{P} = \mathbf{X}^+ \mathbf{X} = \sum_{i=1}^r \mathbf{v}_i \mathbf{v}_i^\top \quad (28.1.1)$$

is the orthogonal projection onto the row space $\mathcal{C}(\mathbf{X}^\top) = \text{span}(\mathbf{v}_1, \dots, \mathbf{v}_r)$. The next theorem collects what is known about $\hat{\boldsymbol{\beta}}^+ = \mathbf{X}^+ \mathbf{y}$ when $p > n$. Part (a) restates earlier results in the form needed here. Parts (b) and (c) show that $\hat{\boldsymbol{\beta}}^+$ is what two common procedures produce when they are pushed to the limit. Part (d) gives its bias and variance.

Theorem 28.1.1 — The minimum-norm interpolator

Let $\mathbf{X}$ be $n \times p$ with rank r and singular values $d_1 \geq \dots \geq d_r > 0$, and let $\hat{\boldsymbol{\beta}}^+ = \mathbf{X}^+ \mathbf{y}$.

- (a) (*Characterization.*) The least squares estimates are the vectors $\hat{\boldsymbol{\beta}}^+ + \mathbf{v}$ with $\mathbf{v} \in \mathcal{N}(\mathbf{X})$. Among them $\hat{\boldsymbol{\beta}}^+$ is the only one in $\mathcal{C}(\mathbf{X}^\top)$, and it has the smallest Euclidean norm. If $r = n$, then $\hat{\boldsymbol{\beta}}^+ = \mathbf{X}^\top (\mathbf{X} \mathbf{X}^\top)^{-1} \mathbf{y}$ and every least squares estimate interpolates: $\mathbf{X} \hat{\boldsymbol{\beta}}^+ = \mathbf{y}$.
- (b) (*Ridgeless limit.*) For $\lambda > 0$ let $\hat{\boldsymbol{\beta}}_\lambda = (\mathbf{X}^\top \mathbf{X} + \lambda \mathbf{I})^{-1} \mathbf{X}^\top \mathbf{y}$ be the ridge estimate. Then $\|\hat{\boldsymbol{\beta}}_\lambda - \hat{\boldsymbol{\beta}}^+\| \leq (\lambda/d_r^2) \|\hat{\boldsymbol{\beta}}^+\|$, so $\hat{\boldsymbol{\beta}}_\lambda \rightarrow \hat{\boldsymbol{\beta}}^+$ as $\lambda \downarrow 0$.
- (c) (*Gradient descent.*) Let $\mathbf{b}_0 \in \mathbb{R}^p$, $0 < \eta < 2/d_1^2$ and $\mathbf{b}_{t+1} = \mathbf{b}_t + \eta \mathbf{X}^\top (\mathbf{y} - \mathbf{X} \mathbf{b}_t)$. Then $\mathbf{b}_t \rightarrow \hat{\boldsymbol{\beta}}^+ + (\mathbf{I} - \mathbf{P}) \mathbf{b}_0$. In particular, gradient descent started at $\mathbf{b}_0 = \mathbf{0}$ converges to $\hat{\boldsymbol{\beta}}^+$.
- (d) (*Bias and variance.*) If $\mathbb{E}(\mathbf{Y}) = \mathbf{X} \boldsymbol{\beta}$ and $\text{Cov}(\mathbf{Y}) = \sigma^2 \mathbf{I}$, then $\mathbb{E}(\mathbf{X}^+ \mathbf{Y}) = \mathbf{P} \boldsymbol{\beta}$ and $\text{Cov}(\mathbf{X}^+ \mathbf{Y}) = \sigma^2 \sum_{i \leq r} d_i^{-2} \mathbf{v}_i \mathbf{v}_i^\top$. For every $\mathbf{a} \in \mathbb{R}^p$,

$$\mathbb{E}(\mathbf{a}^\top \mathbf{X}^+ \mathbf{Y} - \mathbf{a}^\top \boldsymbol{\beta})^2 = (\mathbf{a}^\top (\mathbf{I} - \mathbf{P}) \boldsymbol{\beta})^2 + \sigma^2 \sum_{i=1}^r \frac{(\mathbf{a}^\top \mathbf{v}_i)^2}{d_i^2}, \quad (28.1.2)$$

and $\mathbb{E} \|\mathbf{X}^+ \mathbf{Y} - \boldsymbol{\beta}\|^2 = \|(\mathbf{I} - \mathbf{P}) \boldsymbol{\beta}\|^2 + \sigma^2 \sum_{i \leq r} d_i^{-2}$.

Proof.

- (a) The first two statements are Theorem 10.4.1, and Proposition 1.4.3 applied to the normal equations of Theorem 6.4.2; $\hat{\beta}^+ = \sum_i (\mathbf{u}_i^\top \mathbf{y} / d_i) \mathbf{v}_i$ lies in $\text{span}(\mathbf{v}_1, \dots, \mathbf{v}_r) = \mathcal{C}(\mathbf{X}^\top)$. If $r = n$, then $\mathbf{X}\mathbf{X}^\top$ is a nonsingular $n \times n$ matrix, and $\mathbf{X}^\top(\mathbf{X}\mathbf{X}^\top)^{-1}$ satisfies the four conditions of Definition 1.3.2, so it equals $\mathbf{X}^+$. Moreover $\mathcal{C}(\mathbf{X}) = \mathbb{R}^n$, so the projection onto $\mathcal{C}(\mathbf{X})$ is $\mathbf{I}$ and every least squares fit equals $\mathbf{y}$.
- (b) Write $c_i = \mathbf{u}_i^\top \mathbf{y}$. By Theorem 27.2.2(a), $\hat{\beta}_\lambda = \sum_{i \leq r} d_i c_i \mathbf{v}_i / (d_i^2 + \lambda)$, and Theorem 27.2.2(d) gives the limit $\hat{\beta}^+ = \sum_{i \leq r} (c_i / d_i) \mathbf{v}_i$; only the rate is new. Subtracting, $\hat{\beta}^+ - \hat{\beta}_\lambda = \sum_{i \leq r} \{\lambda / (d_i^2 + \lambda)\} (c_i / d_i) \mathbf{v}_i$, so by orthonormality $\|\hat{\beta}^+ - \hat{\beta}_\lambda\|^2 \leq (\lambda / d_r^2)^2 \sum_i (c_i / d_i)^2 = (\lambda / d_r^2)^2 \|\hat{\beta}^+\|^2$.
- (c) Complete $\mathbf{v}_1, \dots, \mathbf{v}_r$ to an orthonormal basis of $\mathbb{R}^p$ with vectors $\mathbf{v}_{r+1}, \dots, \mathbf{v}_p$ spanning $\mathcal{N}(\mathbf{X})$. Since $\mathbf{X}^\top(\mathbf{y} - \mathbf{X}\mathbf{b}) = \sum_{i \leq r} d_i (c_i - d_i \mathbf{v}_i^\top \mathbf{b}) \mathbf{v}_i$, the coordinates of $\mathbf{b}_t$ evolve separately:

$$\mathbf{v}_i^\top \mathbf{b}_{t+1} = (1 - \eta d_i^2) \mathbf{v}_i^\top \mathbf{b}_t + \eta d_i c_i \quad (i \leq r), \quad \mathbf{v}_i^\top \mathbf{b}_{t+1} = \mathbf{v}_i^\top \mathbf{b}_t \quad (i > r).$$

For $i \leq r$, subtracting the fixed point c_i / d_i gives $\mathbf{v}_i^\top \mathbf{b}_t - c_i / d_i = (1 - \eta d_i^2)^t (\mathbf{v}_i^\top \mathbf{b}_0 - c_i / d_i) \rightarrow 0$, because $0 < \eta d_i^2 < 2$. The coordinates with $i > r$ never move. So $\mathbf{b}_t \rightarrow \sum_{i \leq r} (c_i / d_i) \mathbf{v}_i + \sum_{i > r} (\mathbf{v}_i^\top \mathbf{b}_0) \mathbf{v}_i = \hat{\beta}^+ + (\mathbf{I} - \mathbf{P}) \mathbf{b}_0$.

- (d) $\mathbf{E}(\mathbf{X}^+ \mathbf{Y}) = \mathbf{X}^+ \mathbf{X} \boldsymbol{\beta} = \mathbf{P} \boldsymbol{\beta}$, and $\text{Cov}(\mathbf{X}^+ \mathbf{Y}) = \sigma^2 \mathbf{X}^+ (\mathbf{X}^+)^T = \sigma^2 \sum_i d_i^{-2} \mathbf{v}_i \mathbf{v}_i^\top$ (Theorem 2.2.1). The mean squared error of $\mathbf{a}^\top \mathbf{X}^+ \mathbf{Y}$ is its squared bias $(\mathbf{a}^\top (\mathbf{P} - \mathbf{I}) \boldsymbol{\beta})^2$ plus its variance, which is Eq. 28.1.2. Summing Eq. 28.1.2 over $\mathbf{a} = \mathbf{e}_1, \dots, \mathbf{e}_p$ gives the last formula. ■

Parts (b) and (c) explain why $\hat{\beta}^+$ matters although nobody chooses it on purpose. Gradient iterates started at zero never leave $\mathcal{C}(\mathbf{X}^\top)$, because every gradient $\mathbf{X}^\top(\mathbf{y} - \mathbf{X}\mathbf{b})$ lies in it. Algorithms for large models are typically started near zero and run until the training error vanishes, and in the linear case they return the minimum-norm interpolator. This is called the *implicit regularization* of the algorithm.

```
for lam in [1.0, 0.1, 0.01, 0.001]:
    b_ridge = np.linalg.solve(X.T @ X + lam * np.eye(p), X.T @ y)
    print(f"lambda = {lam:6.3f}    distance to min-norm {np.linalg.norm(b_ridge -
    b_plus):.2e}")

# gradient descent on ||y - X b||^2 / 2 started at zero
step = 1.0 / np.linalg.norm(X, 2) ** 2
b = np.zeros(p)
for t in range(5000):
    b = b + step * X.T @ (y - X @ b)
print("gradient descent: distance to min-norm", np.linalg.norm(b - b_plus))
```

[Run this code online](#)

For the design of [Example \(Two interpolators\)](#) the ridge estimate at $\lambda = 0.01$ is within 0.0132 of $\hat{\beta}^+$, and the distance shrinks in proportion to λ , as part (b) predicts.

In part (d), the bias $(\mathbf{I} - \mathbf{P}) \boldsymbol{\beta}$ is the component of $\boldsymbol{\beta}$ in the null space, which the data cannot

see and the minimum-norm solution sets to zero. The variance is the familiar σ^2/d_i^2 of Theorem 10.4.1, summed over the directions the data do see. So $\mathbf{a}^\top \hat{\boldsymbol{\beta}}^+$ is unbiased for all $\boldsymbol{\beta}$ exactly when $\mathbf{a}^\top \boldsymbol{\beta}$ is estimable (Exercise A2). For any other $\mathbf{a}$, its bias depends on the invisible part of $\boldsymbol{\beta}$ and can be anything.

28.1.3 The risk with random regressors

To say whether interpolation predicts well, the design must be random, since prediction is at new regressor values. The cleanest case has independent standard normal regressors, where both terms of part (d) can be averaged exactly.

Proposition 28.1.2 — Risk under isotropic regressors

Let the rows of $\mathbf{X}$ be independent $N_p(\mathbf{0}, \mathbf{I}_p)$ vectors, let $\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$ with $\boldsymbol{\varepsilon} \sim N_n(\mathbf{0}, \sigma^2 \mathbf{I})$ independent of $\mathbf{X}$, and let $\mathbf{x}_0 \sim N_p(\mathbf{0}, \mathbf{I}_p)$ be independent of both. The prediction risk $\mathcal{R} = E(\mathbf{x}_0^\top \mathbf{X}^+ \mathbf{Y} - \mathbf{x}_0^\top \boldsymbol{\beta})^2$ is

$$\mathcal{R} = \begin{cases} \sigma^2 \frac{p}{n-p-1}, & p \leq n-2, \\ \|\boldsymbol{\beta}\|^2 \left(1 - \frac{n}{p}\right) + \sigma^2 \frac{n}{p-n-1}, & p \geq n+2. \end{cases} \quad (28.1.3)$$

Proof. Given $\mathbf{X}$ and $\mathbf{Y}$, the vector $\hat{\boldsymbol{\beta}}^+ = \mathbf{X}^+ \mathbf{Y}$ is fixed and $E\{(\mathbf{x}_0^\top (\hat{\boldsymbol{\beta}}^+ - \boldsymbol{\beta}))^2 \mid \mathbf{X}, \mathbf{Y}\} = \|\hat{\boldsymbol{\beta}}^+ - \boldsymbol{\beta}\|^2$, because $\text{Cov}(\mathbf{x}_0) = \mathbf{I}$. So $\mathcal{R} = E\|\hat{\boldsymbol{\beta}}^+ - \boldsymbol{\beta}\|^2$, and by Theorem 28.1.1(d) applied conditionally on $\mathbf{X}$,

$$\mathcal{R} = E\|(\mathbf{I} - \mathbf{P})\boldsymbol{\beta}\|^2 + \sigma^2 E \sum_{i=1}^r d_i^{-2}.$$

Case $p \leq n-2$. $\mathbf{X}^\top \mathbf{X} = \sum_i \mathbf{x}_{(i)} \mathbf{x}_{(i)}^\top$ has the Wishart law $W_p(n, \mathbf{I})$ (Definition 14.2.1), so by Lemma 14.2.4 it is nonsingular with probability one and $E(\mathbf{X}^\top \mathbf{X})^{-1} = \mathbf{I}/(n-p-1)$. Then $r = p$, $\mathbf{P} = \mathbf{I}$, the bias term vanishes, and $\sum_i d_i^{-2} = \text{tr}(\mathbf{X}^\top \mathbf{X})^{-1}$ has mean $p/(n-p-1)$.

Case $p \geq n+2$. The p columns of $\mathbf{X}$ are independent $N_n(\mathbf{0}, \mathbf{I}_n)$ vectors, so $\mathbf{X}\mathbf{X}^\top \sim W_n(p, \mathbf{I}_n)$. By Lemma 14.2.4 again, $\mathbf{X}\mathbf{X}^\top$ is nonsingular with probability one, so $r = n$. The numbers d_i^2 are also the eigenvalues of $\mathbf{X}\mathbf{X}^\top$, so $\sum_i d_i^{-2} = \text{tr}(\mathbf{X}\mathbf{X}^\top)^{-1}$, with mean $n/(p-n-1)$. For the bias, let $\mathbf{Q}$ be any orthogonal $p \times p$ matrix. The rows of $\mathbf{X}\mathbf{Q}$ are independent $N_p(\mathbf{0}, \mathbf{Q}^\top \mathbf{Q}) = N_p(\mathbf{0}, \mathbf{I})$ vectors (Theorem 3.3.1), so $\mathbf{X}\mathbf{Q}$ has the same law as $\mathbf{X}$. One checks from Definition 1.3.2 that $(\mathbf{X}\mathbf{Q})^+ = \mathbf{Q}^\top \mathbf{X}^+$, so the row-space projection of $\mathbf{X}\mathbf{Q}$ is $\mathbf{Q}^\top \mathbf{P} \mathbf{Q}$. Hence $\bar{\mathbf{P}} = E\mathbf{P}$ satisfies $\bar{\mathbf{P}} = \mathbf{Q}^\top \bar{\mathbf{P}} \mathbf{Q}$ for every orthogonal $\mathbf{Q}$. Then $\mathbf{u}^\top \bar{\mathbf{P}} \mathbf{u}$ takes the same value c at every unit vector $\mathbf{u}$, since any unit vector is $\mathbf{Q}\mathbf{e}_1$ for some $\mathbf{Q}$. A symmetric matrix whose quadratic form is constant on the unit sphere is $c\mathbf{I}$, by Theorem 1.7.7. Taking traces, $pc = E\text{tr} \bar{\mathbf{P}} = Er = n$. Therefore $E\|(\mathbf{I} - \mathbf{P})\boldsymbol{\beta}\|^2 = \boldsymbol{\beta}^\top (\mathbf{I} - E\mathbf{P})\boldsymbol{\beta} = \|\boldsymbol{\beta}\|^2(1 - n/p)$. ■

Below $p = n$ the fit is unbiased and its variance grows like $p/(n - p - 1)$, which explodes as p approaches n . Above $p = n$ the variance *falls* again, like $n/(p - n - 1)$. The fit has to interpolate n noisy values, and with many more directions available the interpolator can spread the noise over them thinly. The price is the bias: the fit recovers only the fraction n/p of $\|\beta\|^2$. The formulas do not cover $|p - n| \leq 1$.

28.1.4 Double descent

In applications p is usually a choice, such as the number of basis functions. To compare choices of p on one problem, let the response depend on a pool of D features and let the fit use only the first p . The features left out act as extra noise.

Corollary 28.1.3 — Fitting the first p of D features

Let $\mathbf{x} \sim N_D(\mathbf{0}, \mathbf{I}_D)$, $Y = \mathbf{x}^\top \beta + \varepsilon$ with $\varepsilon \sim N(0, \sigma^2)$ independent of $\mathbf{x}$, and let n independent copies be observed. For $p \leq D$, split $\beta = (\beta_{(1)}^\top, \beta_{(2)}^\top)^\top$ into its first p and last $D - p$ entries, write $\omega^2 = \|\beta_{(2)}\|^2$, fit the minimum-norm least squares estimate $\hat{\beta}^+$ (Proposition 6.10.1) to the first p features, and predict with $(\hat{\beta}^+, \mathbf{0}^\top)^\top$. With $\sigma_p^2 = \sigma^2 + \omega^2$, the excess prediction risk at an independent new $\mathbf{x}_0$ is ω^2 plus the right-hand side of Eq. 28.1.3 with $\beta_{(1)}$ in place of β and σ_p^2 in place of σ^2 .

Proof. Split $\mathbf{X} = [\mathbf{X}_1, \mathbf{X}_2]$ in the same way. Then $\mathbf{Y} = \mathbf{X}_1 \beta_{(1)} + (\mathbf{X}_2 \beta_{(2)} + \varepsilon)$. The entries of $\mathbf{X}_2$ are independent of $\mathbf{X}_1$, so the bracket is a $N_n(\mathbf{0}, \sigma_p^2 \mathbf{I})$ vector independent of $\mathbf{X}_1$. Proposition 28.1.2 applies with $\mathbf{X}_1$, $\beta_{(1)}$ and σ_p^2 , and gives $E\|\hat{\beta}^+ - \beta_{(1)}\|^2$. At the new point, the prediction error $\mathbf{x}_{0,1}^\top (\hat{\beta}^+ - \beta_{(1)}) - \mathbf{x}_{0,2}^\top \beta_{(2)}$ is a sum of two uncorrelated terms with mean zero, so its mean square is $E\|\hat{\beta}^+ - \beta_{(1)}\|^2 + \omega^2$. ■

Example. (Two shapes of signal)

Take $n = 50$ observations, a pool of $D = 200$ features, $\sigma^2 = 0.25$ and $\|\beta\|^2 = 1$. Figure 28.1.1 plots the exact risk of Corollary 28.1.3 against p for two coefficient vectors, with simulated values as a check. It also shows the risk of ridge regression (Theorem 27.2.2) with the best penalty for each p .

In panel (a) all 200 coefficients are equal, so each feature left out adds to the noise. No model with $p < n$ beats predicting zero, which has risk 1.000. The risk climbs to 12.31 at $p = 45$, is 12.94 at $p = 55$, and then falls. The overall minimum is at $p = 200$: 0.834. Here the best prediction comes from interpolating with every feature.

In panel (b) the coefficients decay geometrically, and the classical picture holds: a small model with $p = 9$ has risk 0.078, and nothing to the right of the peak comes close. At $p = D$ the risk is 0.834 again. When all features are used, Eq. 28.1.3 depends on β only through $\|\beta\|$.

A risk curve that falls, peaks at $p = n$ and falls again is called **double descent**. Three cautions

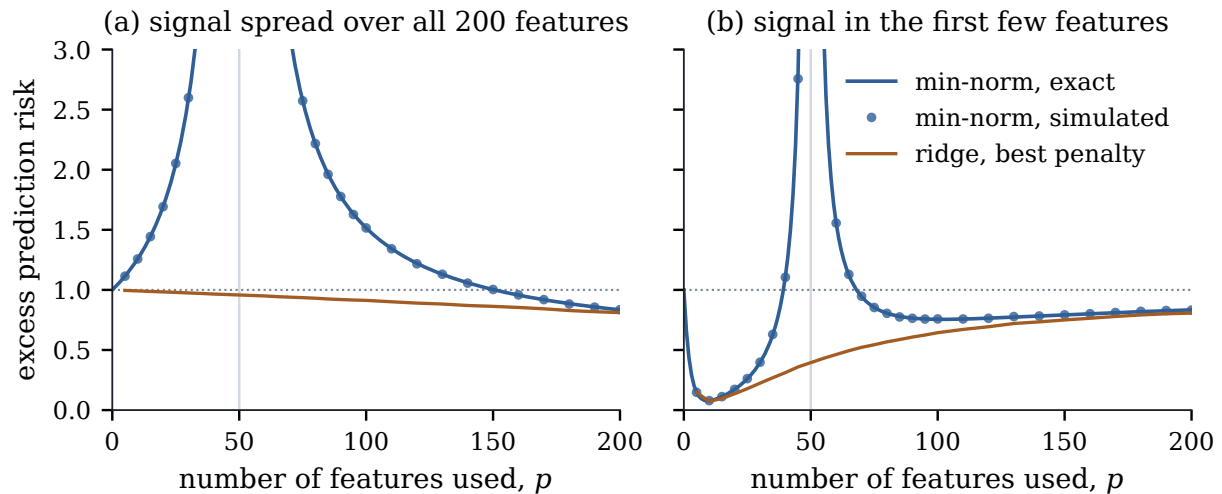


Figure 28.1.1: Excess prediction risk of the minimum-norm least squares fit on the first p of $D = 200$ isotropic features, with $n = 50$ (vertical line) and $\sigma^2 = 0.25$: the exact formula of [Corollary 28.1.3](#) (line) and simulation (dots), and ridge regression with the best penalty for each p . The dotted line is the risk of predicting zero. (a) Equal coefficients. (b) Geometrically decaying coefficients.

keep it in proportion.

- *The peak belongs to the estimator, not to the problem.* It comes from the variance term $\sigma^2 \sum_i d_i^{-2}$, large when the smallest nonzero singular value is near zero, as for $p \approx n$. Ridge with the best penalty has no peak, and its best values, 0.811 and 0.073, are slightly below the best minimum-norm values, as [Theorem 28.1.1\(b\)](#) requires.
- *The second descent need not win.* That depends on how the signal is spread, as the panels show.
- *Isotropic features are a special case.* Hastie, Montanari, Rosset and Tibshirani (2022) treat general covariances as $n, p \rightarrow \infty$ with p/n fixed, and Bartlett, Long, Lugosi and Tsigler (2020) find when interpolation is *benign*, with risk still tending to that of the best predictor.

```
import numpy as np

def exact_risk(p, n, beta, sigma2):
    """Excess prediction risk of min-norm least squares on the first p of D
    isotropic features."""
    kept, left = np.sum(beta[:p] ** 2), np.sum(beta[p:] ** 2)
    noise = sigma2 + left # omitted features act as extra noise
    if p == 0:
        return left
    if p <= n - 2:
        return left + noise * p / (n - p - 1)
    if p >= n + 2:
        return left + kept * (1 - n / p) + noise * n / (p - n - 1)
    return np.inf # |p - n| <= 1: not covered by the formula

def simulated_risk(p, n, beta, sigma2, reps, rng, lams=()):
    """Average over designs of the conditional risk given X; also ridge for each
    penalty."""
```

```

bS, left = beta[:p], np.sum(beta[p:] ** 2)
noise = sigma2 + left
out, ridge = [], np.zeros(len(lams))
for _ in range(reps):
    X = rng.normal(size=(n, p))
    _, d, Vt = np.linalg.svd(X, full_matrices=False)
    c = Vt @ bS                                # coordinates of beta_S in the
row space
    bias2 = bS @ bS - c @ c                    # ||(I - P) beta_S||^2
    out.append(left + bias2 + noise * np.sum(1 / d ** 2))
    if len(lams):
        shrink = d ** 2 / (d ** 2 + np.asarray(lams)[: , None])    # one row per
penalty
        ridge += left + bias2 + np.sum(((1 - shrink) * c) ** 2, axis=1) \
            + noise * np.sum(shrink ** 2 / d ** 2, axis=1)
return np.mean(out), np.std(out) / np.sqrt(reps), ridge / reps

n, D, sigma2 = 20, 100, 0.25
beta = np.full(D, 1 / np.sqrt(D))            # signal spread evenly, ||beta
||^2 = 1
rng = np.random.default_rng(2802)
for p in [5, 10, 15, 25, 40, 100]:
    sim, se, _ = simulated_risk(p, n, beta, sigma2, reps=200, rng=rng)
    print(f"p = {p:3d}: formula {exact_risk(p, n, beta, sigma2):.3f}    simulation {
sim:.3f}")

```

[Run this code online](#)

For the rest of the chapter we take a different route. Instead of letting the algorithm decide which of the many solutions to report, we *assume* that the truth is simple: only a few coefficients are nonzero. That assumption is strong, but it is explicit, and it can be exploited through the lasso.

28.1.5 Exercises

28.1.5.1 A. Check your understanding

Exercise A1.

Let $\mathbf{X}$ be $n \times p$ with an intercept column and $\text{rank}(\mathbf{X}) = n < p$. Show that every least squares fit has $R^2 = 1$, and explain why neither the F test of Theorem 11.1.1 nor the residual-based diagnostics of Chapter 20 can be used. What are the leverages h_{ii} ?

Exercise A2.

In the setting of Theorem 28.1.1(d), show that $\mathbf{a}^\top \mathbf{X}^+ \mathbf{Y}$ is unbiased for $\mathbf{a}^\top \boldsymbol{\beta}$ for every $\boldsymbol{\beta}$ iff $\mathbf{a} \in \mathcal{C}(\mathbf{X}^\top)$. Show also that for such $\mathbf{a}$ its variance is $\sigma^2 \mathbf{a}^\top (\mathbf{X}^\top \mathbf{X})^+ \mathbf{a}$, and that it equals the least squares estimator $\mathbf{a}^\top \hat{\boldsymbol{\beta}}$ of Theorem 6.4.4 for any least squares estimate $\hat{\boldsymbol{\beta}}$.

Solution. The bias is $\mathbf{a}^\top (\mathbf{P} - \mathbf{I}) \boldsymbol{\beta}$, which vanishes for all $\boldsymbol{\beta}$ iff $(\mathbf{I} - \mathbf{P}) \mathbf{a} = \mathbf{0}$, that is, iff $\mathbf{a} \in \mathcal{C}(\mathbf{P}) = \mathcal{C}(\mathbf{X}^\top)$. The variance is $\sigma^2 \sum_i (\mathbf{a}^\top \mathbf{v}_i)^2 / d_i^2$, and $\sum_i d_i^{-2} \mathbf{v}_i \mathbf{v}_i^\top$ is the

Moore–Penrose inverse of $\mathbf{X}^\top \mathbf{X} = \sum_i d_i^2 \mathbf{v}_i \mathbf{v}_i^\top$ (Theorem 1.8.1). Finally, any least squares estimate is $\hat{\boldsymbol{\beta}}^+ + \mathbf{v}$ with $\mathbf{v} \in \mathcal{N}(\mathbf{X})$, and $\mathbf{a}^\top \mathbf{v} = 0$ because $\mathbf{a} \in \mathcal{C}(\mathbf{X}^\top) = \mathcal{N}(\mathbf{X})^\perp$, so $\mathbf{a}^\top \hat{\boldsymbol{\beta}} = \mathbf{a}^\top \hat{\boldsymbol{\beta}}^+$.

28.1.5.2 B. Practice

Exercise B1.

Show directly from the update in Theorem 28.1.1(c) that $(\mathbf{I} - \mathbf{P})\mathbf{b}_t = (\mathbf{I} - \mathbf{P})\mathbf{b}_0$ for all t . Deduce that gradient descent started at $\mathbf{b}_0$ returns the least squares estimate closest to $\mathbf{b}_0$.

Exercise B2.

Fix $\mathbf{X}$ of rank r and let $m(\lambda) = \mathbb{E}\|\hat{\boldsymbol{\beta}}_\lambda - \boldsymbol{\beta}\|^2$ for the ridge estimate, with $m(0)$ the risk of $\hat{\boldsymbol{\beta}}^+$ in Theorem 28.1.1(d). Show that

$$m(\lambda) = \|(\mathbf{I} - \mathbf{P})\boldsymbol{\beta}\|^2 + \sum_{i=1}^r \left[\left(\frac{\lambda}{d_i^2 + \lambda} \right)^2 (\mathbf{v}_i^\top \boldsymbol{\beta})^2 + \sigma^2 \frac{d_i^2}{(d_i^2 + \lambda)^2} \right],$$

and that $m'(0) < 0$ whenever $\sigma > 0$. So some ridge estimate beats the minimum-norm interpolator for every $\boldsymbol{\beta}$. Compare Theorem 27.2.3.

Solution. The linear representation in Theorem 27.2.2(a) holds for any rank r . Applied to $\mathbb{E}(\mathbf{Y}) = \mathbf{X}\boldsymbol{\beta}$, whose coordinates are $\mathbf{u}_i^\top \mathbf{X}\boldsymbol{\beta} = d_i \mathbf{v}_i^\top \boldsymbol{\beta}$, it gives $\mathbb{E} \hat{\boldsymbol{\beta}}_\lambda = \sum_{i \leq r} \{d_i^2 / (d_i^2 + \lambda)\} (\mathbf{v}_i^\top \boldsymbol{\beta}) \mathbf{v}_i$, and the covariance is $\sigma^2 \sum_{i \leq r} d_i^2 (d_i^2 + \lambda)^{-2} \mathbf{v}_i \mathbf{v}_i^\top$, as in Theorem 27.2.2(c). What $r < p$ adds is the bias component $-(\mathbf{I} - \mathbf{P})\boldsymbol{\beta}$ in the null space, orthogonal to the rest of the bias $-\sum_i \{\lambda / (d_i^2 + \lambda)\} (\mathbf{v}_i^\top \boldsymbol{\beta}) \mathbf{v}_i$; it contributes $\|(\mathbf{I} - \mathbf{P})\boldsymbol{\beta}\|^2$. Adding the squared bias and the trace of the covariance gives $m(\lambda)$. Differentiating term by term, the first bracket has derivative $2\lambda d_i^2 (\mathbf{v}_i^\top \boldsymbol{\beta})^2 / (d_i^2 + \lambda)^3$, which is zero at $\lambda = 0$. The second has derivative $-2\sigma^2 d_i^2 / (d_i^2 + \lambda)^3$, which equals $-2\sigma^2 / d_i^4 < 0$ at $\lambda = 0$. Hence $m'(0) = -2\sigma^2 \sum_i d_i^{-4} < 0$, and $m(\lambda) < m(0)$ for small $\lambda > 0$.

28.1.5.3 C. Going deeper

Exercise C1.

Check Eq. 28.1.3 for $n = 1$ directly. With one observation $(\mathbf{x}, y)$, show that $\hat{\boldsymbol{\beta}}^+ = \mathbf{x}y / \|\mathbf{x}\|^2$, and use $\mathbb{E}(1/\chi^2(p)) = 1/(p-2)$ for $p \geq 3$ to show that $\mathcal{R} = \|\boldsymbol{\beta}\|^2(1 - 1/p) + \sigma^2/(p-2)$.

Solution. Here $\mathbf{X} = \mathbf{x}^\top$ and $\mathbf{X}\mathbf{X}^\top = \|\mathbf{x}\|^2$, so $\hat{\boldsymbol{\beta}}^+ = \mathbf{x}y / \|\mathbf{x}\|^2$ and $\hat{\boldsymbol{\beta}}^+ - \boldsymbol{\beta} = -(\mathbf{I} - \mathbf{x}\mathbf{x}^\top / \|\mathbf{x}\|^2)\boldsymbol{\beta} + \mathbf{x}\varepsilon / \|\mathbf{x}\|^2$, a sum of orthogonal vectors. So

$$\|\hat{\boldsymbol{\beta}}^+ - \boldsymbol{\beta}\|^2 = \|\boldsymbol{\beta}\|^2 - \frac{(\mathbf{x}^\top \boldsymbol{\beta})^2}{\|\mathbf{x}\|^2} + \frac{\varepsilon^2}{\|\mathbf{x}\|^2}.$$

The matrix $\mathbf{P} = \mathbf{x}\mathbf{x}^\top / \|\mathbf{x}\|^2$ is the row-space projection, and the rotation argument in the proof of Proposition 28.1.2 gives $\mathbb{E} \mathbf{P} = \mathbf{I}/p$, so $\mathbb{E}(\mathbf{x}^\top \boldsymbol{\beta})^2 / \|\mathbf{x}\|^2 = \boldsymbol{\beta}^\top (\mathbb{E} \mathbf{P}) \boldsymbol{\beta} = \|\boldsymbol{\beta}\|^2 / p$.

And $\|\mathbf{x}\|^2 \sim \chi^2(p)$ is independent of ε , so $E \varepsilon^2 / \|\mathbf{x}\|^2 = \sigma^2 / (p - 2)$. This agrees with [Eq. 28.1.3](#) at $n = 1$.

Exercise C2.

In [Proposition 28.1.2](#) let $n, p \rightarrow \infty$ with $n/p \rightarrow \gamma \in (0, 1)$ and $\|\boldsymbol{\beta}\|^2$ fixed. Show that $\mathcal{R} \rightarrow \|\boldsymbol{\beta}\|^2(1 - \gamma) + \sigma^2\gamma/(1 - \gamma)$, and that this is below the risk $\|\boldsymbol{\beta}\|^2$ of predicting zero iff $\|\boldsymbol{\beta}\|^2/\sigma^2 > 1/(1 - \gamma)$. What happens to the limit as $\gamma \rightarrow 0$?